

SAKET SANKHLA

📍 Jodhpur, Rajasthan, India

☎ +91 7891093675

saketsankh18@gmail.com

linkedin.com/in/saket-sankhla-4a193a277

github.com/SaketSankhla

EDUCATION

Government Engineering College, Ajmer (GECA) — Ajmer, Rajasthan, India

Sept 2023 – June 2027 (Expected)

Bachelor of Technology (B.Tech) in Electronics & Communication Engineering

BTU Affiliate

- Key Coursework: Digital System Design, Linear Integrated Circuits, Computer Architecture, Optical Fiber Communication, Neural Networks & Fuzzy Logic.

KL University — Guntur, Andhra Pradesh, India

July 2023 – Sept 2023

Bachelor of Technology (B.Tech) in Electronics & Communication Engineering (Transferred)

TECHNICAL SKILLS

VLSI & Digital Systems	RTL Design, Verilog, SystemVerilog, VHDL, APB & AHB Bus Protocols, Digital Circuit Design, Timing & Waveform Analysis
EDA Tools & Platforms	Xilinx Vivado, ModelSim, GTKWave, Yosys, OpenLane, Icarus Verilog, Altium Designer
Embedded & IoT	ESP32, Arduino, Microcontroller Architecture, PCB Layout & Schematics, Sensor Integration, I2C, SPI, UART
Programming Languages	C, C++ (Embedded), Python, Assembly (8085/8086), Dart, JavaScript

EXPERIENCE

Defence Research and Development Laboratory (DRDL) - DRDO — Jodhpur, India

May 2026 – Present

Summer Intern

- Analyze core communication electronics and RF module integrations for defense applications.
- Participate in hardware testing, debugging, and quality-assurance evaluation of avionics control electronics.
- Document system configurations and electrical specifications for high-reliability embedded hardware.

IIT Roorkee Drone Hackathon — Roorkee, India

2026

Electronics Lead

- Designed custom power distribution logic and sensor-feedback control circuits for autonomous flight nodes.
- Resolved flight stabilization issues by tuning sensor integration loops and low-level firmware parameters.
- Collaborated with mechanical and software leads to verify system integration under tight competition timeline constraints.

Maven Silicon — *Bengaluru, India*

July 2025 – August 2025

VLSI Design Intern

- Modeled and simulated a chip-level APB-to-AHB communication protocol bridge in SystemVerilog.
- Performed RTL simulation and debugging using Xilinx Vivado, checking signal transitions and pipeline register transfers.
- Conducted timing and waveform analysis, locating and eliminating setup/hold time interface violations to ensure protocol compliance.

Bharat Sanchar Nigam Limited (BSNL) — *Jodhpur, India*

June 2024

Telecom Trainee

- Studied the operational infrastructure of GSM, PSTN switching systems, and optical fiber transport networks.
- Analyzed routing, transmission interfaces, and physical cable termination configurations at central exchange stations.

PROJECTS

Electricity Theft Detector v2.0 — *ESP32, Python, C++, TensorFlow*

- Engineered an ESP32-based smart energy meter utilizing ZMPT101B voltage and SCT-013 current sensors to measure grid loads.
- Programmed C++ firmware implementing True RMS calculations and a 3-window debounce algorithm, reducing false theft alarms by 40%.
- Configured a Python data pipeline to train CNN-LSTM classifiers on the SGCC smart grid dataset, enabling high-accuracy detection of non-technical distribution losses (theft).
- Integrated Google Sheets API logging and established automatic reconnect mechanisms with watchdog hardware timers.

Arduino JARVIS Desk Companion — *Arduino, Embedded C, Sensors*

- Developed an interactive desk assistant integrating HC-SR04 sonar detection, a DHT11 environmental sensor, and an I2C 1602 LCD display.
- Structured a non-blocking task scheduler in Embedded C to poll sensors and drive display updates, maintaining low-latency responses.
- Designed custom physical wiring configurations and noise-filtration components for stable sensor signal lines.

Digital VLSI Library Blocks — *Verilog, Icarus Verilog, GTKWave*

- Wrote synthesizable Verilog modules for fundamental arithmetic and combinational circuits, including 4-bit Full Adders (fa_4bit), Decoders (2-to-4), and multiplexer configurations (2-to-1, 4-to-1, 8-to-1).
- Verified logical correctness using custom testbenches, running simulations on Icarus Verilog and analyzing timing signals in GTKWave.